Task Recognition from Physiological Data Using Multimodal Sensor Fusion for Human-Robot Collaboration

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Abstract: Task-type recognition using physiological information of operators is an essential capability in adaptive and assistive human-robot cooperation. Appropriate assistance can be adapted in real time by enabling a robot to detect an operator's current task state from physiological signals quickly and reliably. This requires handling differences between operators, changes in sensor relationships over time, and imbalanced label distributions. We propose, in this work, a multimodal pipeline for uniform pre-processing and fusion of features from sensors to a tabular format for training detectors with class-based objectives. The pipeline is applied to MultiPhysio-HRC dataset. Different task-types are recognized as cognitive load, high stress, industrial task, low load, and other. Results show that the fused representation, with XGBoost achieves the best performance with 0.95 accuracy and 0.91 macro-F1. A modality ablation confirms that the fused input improves macro-F1 over the single modality for every detector tested, by 0.047 to 0.186.

1 INTRODUCTION

Human-Robot Collaboration (HRC) is a paradigm of shared autonomy in which a human operator and a robot cooperate to accomplish a certain task in an efficient and safe way (Selvaggio et al., 2021; Borghi et al., 2025). For that, a robot needs to be able to constantly recognize the state of an operator and adapt its behavior accordingly to the progress of the interaction. The implementation of this feature involves a training process based on the data collected using sensors from the human operator and the robot. The data of the human side includes physiological and behavioral features, such as Electroencephalography (EEG), Electromyography (EMG), Electrodermal Activity (EDA), Electrocardiography (ECG), respiration and others taken from multimodal dataset designed to discover the intentions of the person.

Although different types of sensors provide different data, the problem of poor fusion leads to overfitting of the subject-specific data or absence of intermodal correlations. Moreover, labels for target tasks are often unbalanced and can not be perfectly separable from the current representation, leading to overfitting and unreliable predictions.

In this paper, we propose a system designed for task-condition recognition, which is based on MultiPhysio-HRC multimodal dataset containing five task conditions: cognitive load, high stress, industrial task, low load and other (Bussolan et al., 2025b). We build a compact multimodal fusion pipeline that maps heterogeneous physiological signals to a single tabular representation, compare five detectors under one shared protocol, and test whether fusing modalities is justified rather than assuming it. We report the resulting baselines together with their scope so that they can be used in assistive HRC applications.

The rest of the paper is structured as follows: Section 2 reviews related work on physiological task recognition and multimodal fusion in HRC. Section 3 describes the proposed methodology, covering the pipeline, the data and the label schema, the problem formulation, detector selection, the training process, and the metrics. Section 4 presents the benchmark, the modality ablation, the class-level analysis, and the latency measurements, and Section 5 concludes.

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2 RELATED WORK

Physiological operator-state recognition for HRC has advanced through new datasets, new fusion models, and new evaluation studies. This section outlines recent contributions in each of these areas and identifies the drawback that motivates the present work.

The MultiPhysio-HRC dataset used in this paper was introduced by Bussolan et al. (2025b), who reported recognition of self-reported stress and cognitive-load levels on an industrial assembly task, reaching macro-F1 ranges of roughly 0.34 to 0.39 for stress and 0.33 to 0.49 for cognitive load across conditions (Bussolan et al., 2024). Those targets come from questionnaire instruments that are not part of the public feature export, and the source study neither audits its feature export for leakage nor tests whether fusing modalities helps under a controlled design. Our work employs the same dataset but another protocol-based target alongside the audit and fusion test.

Borghi et al. (2025) studied the stress level of operators during collaboration with robots via a multimodal approach and found that there is considerable variability across individuals. Aygun et al. (2022), who evaluated cognitive workload in assistive robots through physiological data, found significant variability among individuals as well. All these studies indicate the fact that the operator effect is one of the key challenges, yet all of them look at a single target state in a single architecture pipeline.

On the fusion side, Baltrušaitis et al. (2019) provide the standard taxonomy of multimodal machine learning, arguing that heterogeneous streams carry complementary information and that naive fusion can hurt when modalities are redundant or noisy. Recent work has begun to pair fusion with explicit subject-independent design, as in Cheng et al. (2025), who fuse EEG with eye-tracking for cross-subject cognitive-load recognition and combine a cross-attention fusion mechanism with domain-adversarial adaptation so that the learned representation transfers to unseen participants. That combination is promising, but it still assumes rather than tests the value of fusion itself. The taxonomy of Baltrušaitis et al. (2019) remains largely descriptive, and controlled empirical tests on physiological HRC data, where redundancy across channels is high, remain scarce. Our modality ablation is a step toward filling that gap.

On the evaluation side, Iarlori et al. (2024) survey cognitive-workload estimation from wearable sensors in human-machine interaction and catalogue the evaluation heterogeneity that makes studies hard to compare. Jin et al. (2025) provide a recent systematic review of physiological cognitive-state recognition and show that the design of the data split is among the largest drivers of reported performance differences. That concern has since been quantified: Brookshire et al. (2024) contrast segment-based and subject-based holdouts on EEG and find that the segment-based protocol strongly overestimates performance on new subjects, and their survey of the literature shows that most translational deep-learning studies use exactly that protocol. Reviews of the workload literature reach a parallel conclusion from the task side, where Pušica et al. (2025) report a significant drop in classification accuracy when workload is induced by multitasking and rated by quantitative task load rather than subjectively, and Diarra et al. (2025) note that the metrics used in controlled laboratory studies rarely transfer to real-world settings. These reviews identify the problem but do not supply a concrete recipe for the compact applied setting we target.

Selvaggio et al. (2021) survey autonomy levels in physical human-robot interaction and identify reliability, transparency, and adaptation as the constraints that any state estimator must serve. Task-state recognition is a building block of such autonomy, and our latency and failure-mode discussion speaks to the deployment requirements they enumerate.

Recent work points toward several responses to the difficulties above. Bussolan et al. (2025a) extend the same line of multimodal operator monitoring with federated learning, training on-device over EEG, ECG, EDA, EMG, and respiration so that stress prediction is personalized without moving raw signals off the operator's equipment. Wei et al. (2025) address the task-dependence of workload signatures instead, using a stacked graph attention convolutional network that reaches 0.651 accuracy when the model must generalize across N-back, mental arithmetic, and Sternberg tasks, which shows how much accuracy is lost once the task itself no longer identifies the label. At the representation level, Jiang et al. (2024) pretrain a large EEG model on roughly 2,500 hours drawn from about 20 datasets and fine-tune it across clinical and affective tasks, which shows that generic EEG representations are now feasible. None of these lines, however, reports a compact audited pipeline on a single public HRC dataset that separates the contribution of the features from the contribution of fusion, which is the gap we address.

In summary, prior work establishes that fusion of physiological modalities is promising in principle (Baltrušaitis et al., 2019; Cheng et al., 2025), that operator-state recognition in HRC is hard and subject-dependent, and that evaluation design determines reported performance. What is missing is a compact

and reproducible account of task-condition recognition on a public HRC dataset that audits the features before modelling, states the label rule exactly, and tests whether fusion helps. The present work addresses that gap.

3 METHODOLOGY

3.1 Proposed Solution Pipeline

This solution is implemented based on the pipeline described in Figure 1. First, the physiological signals get processed and normalized in pre-processing phase, and then get converted into a tabulated form. Alongside that fused representation we retain two modality feature groups, one holding the EEG channels and one holding the peripheral channels, because the ablation of Section 4.2 reuses the identical preprocessing path on each group separately. The three representations are passed to the selected detector models, which ensures a consistent input structure and therefore makes any difference in outcome attributable to the detector rather than to the data preparation.

The preprocessing stage carries more weight than its name suggests, because the export it consumes is not yet a clean feature matrix. That export is a table of 257 columns in which physiological summaries sit beside the acquisition pipeline's own bookkeeping, namely operator and repetition identifiers, dataset flags, and duplicate copies of columns that already exist. Removing that bookkeeping belongs to the method itself rather than to housekeeping, since a model trained on it learns the recording instead of the task and counts some evidence twice, and a pipeline that skips this step cannot separate what the model learned about the task from what it learned about the recording. Section 3.2 states the audit in full, and every detector reported below is trained on the 128 features that survive it.

The fusion step itself operates as follows. The EEG block contributes 96 features, which consist of 12 summary statistics per eight channels. Peripheral block contributes with 32 features, which are 8 summary statistics per four signal families (ECG, EDA, EMG and respiration). The combination of the two blocks leads to a 128-dimensional fused vector, which serves as an input to every tabular classifier. In the two-tower deep learning model, the two blocks are not combined on the input layer. Rather, every block is projected into the small embedding by the corresponding tower, and then two embeddings are combined before the output layer. This design lets the network learn a separate representation per modality group before mixing them, which is the structural difference between the two-tower model and the tabular detectors that receive the same 128 features as one flat vector.

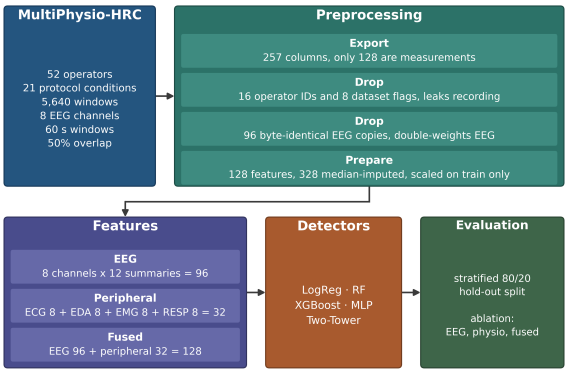


Figure 1: Proposed pipeline, from preprocessing to the shared evaluation protocol.

3.2 Data and Preprocessing

The successful implementation of the detector requires only one preprocessing workflow that is consistent for all models. The main training data set comes from MultiPhysio-HRC, which is a multimodal industrial HRC data set with combined physiological and behavioral measurements in both controlled and realistic settings (Bussolan et al., 2025b). Thus, the same preprocessing steps have been used for each data set. First, all metadata and non-features are deleted. All candidate features then get converted into numerical format, and any columns that are null for the whole export are dropped. Task labels are then mapped into the unified five-class schema of Table 1. Missing values are filled by median imputation, the features are standardized using training-fold statistics only, and the result is assembled into the unified tabular fused representation together with the per-modality feature groups that the ablation of Section 4.2 reuses.

The column audit deserves a precise statement, because the export we obtained is wider than the physiological measurement it is supposed to contain. Of its 257 columns only 128 hold a genuine physiological feature, and the remaining 129 are bookkeeping of four different kinds that must be separated rather than treated alike. 16 columns carry recording identifiers and trial counters, among them columns prefixed `ID__` and `Repetition__`, which record which operator and which repetition produced the row. 8 further columns are dataset metadata and presence flags,

which describe the export rather than the person. Neither of the two families above includes a measure that is related to the operator.

Identifier column is an explicitly defined path through which the classifier learns to detect the recording and not the actual class (Brookshire et al., 2024). This needs to be described clearly since in one recording, the identity of the operator does not change while that of the class does. An algorithm able to parse the identifier will be able to remember the identity of the operator of a particular window and score well on withheld data for that operator without having learned any task relevant information. We drop these 24 columns, and with them 24 opportunities to memorize the recording.

96 further columns are byte-identical duplicates, meaning that each one repeats the exact numerical values of an existing EEG feature column under a different name. They add no information, but they are not harmless either. A model that weighs columns as evidence now receives the EEG block twice, so the duplication would silently give EEG a weight of 192 columns against 32 peripheral columns, and any comparison between modalities would be settled by the bookkeeping rather than by the signal. We drop these 96 columns as well, which brings the number of removed columns to 120.

The remaining nine columns are identifier and label keys. Unlike the columns above they are not deleted, because the split and the scoring still need to know which window belongs to which class. They are held aside and are never passed to a detector as inputs. What the detectors see is therefore the 128 genuine features, distributed as 96 EEG, 8 ECG, 8 EDA, 8 EMG, and 8 respiration features, which is 257 minus the 120 removed columns and the 9 keys. The retained set contains 328 missing values in total, which median imputation fills using training-fold statistics only.

3.3 Task-Condition Labels

The dataset annotates each recording with the protocol condition that was performed. We therefore define the recognition target as the task condition, which the experimental protocol itself certifies. Each window receives exactly one of five labels, assigned deterministically from the recording's condition name by the rule released with the reproducibility package. Because the labels are a deterministic function of the protocol, they are exact and reproducible, and there is no ambiguity about which condition a window came from.

The five classes are defined by the protocol conditions that produce them, rather than by an abstract psychological interpretation. Table 1 gives the complete mapping for all 21 conditions together with the resulting window counts.

The mapping is coherent with a physiological hypothesis in each case. The cognitive-load conditions, namely the Stroop battery in its easy and hard variants, the N-back task, the Tower of Hanoi, and an arithmetic and puzzle task, tax working memory and executive control. High stress involves a virtual reality height plank walking paradigm, which is a verified stressor that causes autonomic arousal due to the perception of threat. Industrial task involves ten blocks of physical assembly and order picking, out of which five blocks are done without the robot while five others are completed with a collaborative robot. Low load involves rest and guided meditation segments inserted between high demand tasks as a low-demand baseline. The residual class holds a single open-ended virtual-reality job-simulation session that belongs to none of the controlled batteries. We keep it as its own class rather than forcing it into another, and we report its small size explicitly.

The class distribution is strongly imbalanced, with 3,571 windows for industrial task against 224 for high stress and 71 for other. In proportional terms the majority class accounts for 63.3 percent of the data, high stress for 4.0 percent, and other for 1.3 percent. A classifier that always predicts industrial task therefore reaches 0.633 accuracy while recovering none of the minority states, which is why accuracy alone cannot rank models on this dataset and why the metrics of Section 3.7 and the class weights of Section 3.4 are necessary.

3.4 Problem Formulation

Task recognition using physiological signals in this work is a supervised classification problem that involves five classes. Let z_i be the measurements derived from the operator through the different sensors synchronized for a particular observation period. After the preprocessing and fusion, each observation period is represented in terms of a feature vector $x_i \in \mathbb{R}^d$ with $d = 128$ along with its corresponding class $y_i \in \{1, \ldots, 5\}$ representing the five task conditions in the order of cognitive load, high stress, industrial task, low load, and other.

Given a model f_θ with parameters θ, the class posterior for class c on input x_i is written $p_\theta(y = c \mid x_i)$. To handle the class imbalance documented above, training minimizes a class-weighted cross-

Table 1: Deterministic mapping from the 21 protocol conditions to the five classes.

Class	Protocol conditions, `task_name`	Windows
cognitive load	`stroopeasy_0`, `stroophard_0`, `n-back_0`, `hanoi_0`, `mat_0`	975
high stress	`vr-plank_0`	224
industrial task	`manual-task-1...-5`, `cobot-task-1...-5`	3,571
low load	`rest-1`, `rest-2`, `rest_0`, `meditation_0`	799
other	`vr-job-sim_0`	71
Total	21 conditions	5,640

entropy objective:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} w_c \,\mathbf{1}[y_i = c]\log p_\theta(y = c \mid x_i), \quad (1)$$

where N is the number of training windows, c indexes the five classes, $\mathbf{1}[\cdot]$ is the indicator function, and w_c is the weight assigned to class c. The predicted label is obtained by

$$\hat{y}_i = \arg\max_{c\in\{1,\ldots,5\}} p_\theta(y = c \mid x_i). \quad (2)$$

The weight w_c is the part of Eq. (1) that responds to the measured imbalance. We set $w_c = N/(5N_c)$ and then rescale the vector so that its mean is one, where N_c is the number of training windows of class c. With equal class sizes this reduces to uniform weighting. On the observed distribution it assigns the majority class a weight of about 0.32 and the smallest class a weight of about 16, which is what prevents the objective from being minimized by a model that simply reproduces the majority class. The word *class-aware* is used throughout to mean exactly this weighting scheme.

3.5 Algorithm Selection

For this research, a collection of algorithms considered practical for application in fused tabular data models will be used: logistic regression, random forest, eXtreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), Multilayer Perceptron (MLP), and a simple two-tower fusion architecture. The two-tower approach employs the explicit separation of modality groups, as described in Section 3. Thus, the emphasis is put on the operational compact fusion design.

3.6 Training Procedure

The training process is meant to be simple, clear, and reproducible for all the analyzed algorithms: first of all, an 80/20 train-test split is done in a stratified way with the random seed set to 42, the preprocessing is fit only to the training data, then applied to the test data. For neural networks, the class-weighted cross-entropy loss with Adam optimizer is used with early stopping applied with the fixed seed and stratified by class on the early-stopping subset; for classic models, the class-balancing is performed by weighting the models where applicable. This training protocol guarantees that any performance differences observed will be a result of different models' behavior, not data preprocessing specifics. The hyperparameters settings are given in Table 2.

3.7 Metrics and Diagnostics

For the current five-class physiological data set, the choice of metrics is affected by the imbalance between the classes and the need for consistent performance throughout all task states, according to the literature (Sokolova and Lapalme, 2009). The accuracy metric is proposed as an overall indicator of correctness of classification results, but is not considered conclusive. Section 3 reveals that the classification accuracy achieved based only on the majority class is 0.633, and previous studies show that the accuracy might be a misleading metric in case of this degree of class imbalance (Jeni et al., 2013). Thus, balanced accuracy, representing the average of the recalls of all the classes and better representing the sensitivity to the minority classes such as the low-load task and other similar task states, is included (Brodersen et al., 2010). Furthermore, macro-F1 is chosen as the main ranking metric due to its penalty on low precision and recall of each class equally weighting all classes.

In addition to the scalar metrics, confusion matrix and per-class precision, recall and F1 are analyzed for the selected classifier. This helps to understand the pairs of tasks being confused and makes sure that robust aggregate performance is achieved for each class separately using the same stratified 80/20 hold-out procedure.

Table 2: Detector hyperparameters on the shared 128-dimensional representation.

Model	Configuration
Logistic Regression	L2 penalty, `max_iter` 2000, balanced class weights
Random Forest	300 trees, balanced class weights
XGBoost	400 estimators, depth 6, learning rate 0.05, subsample 0.9, column subsample 0.9
MLP	hidden layers 128 and 64, Adam, `max_iter` 600
Two-Tower Fusion	per-modality towers 128 then 64, head 64 then 5, dropout 0.2, Adam at 1e-3, weight decay 1e-4, batch 256, early-stopping patience 8

4 RESULTS AND DISCUSSION

4.1 Comparison

We compare the proposed detector against the selected baselines under the stratified 80/20 hold-out and identify the strongest operating point for the fused physiological representation. Table 3 summarizes the resulting ranking, and Figure 2 shows the same comparison as grouped bars so that the three metrics can be read side by side.

Table 3: Benchmark results under the stratified 80/20 hold-out, ranked by macro-F1.

Model	Accuracy	Bal. Acc.	Macro-F1
XGBoost	0.9512	0.8819	0.9104
Random Forest	0.9264	0.8350	0.8773
MLP	0.9184	0.8391	0.8457
Two-Tower Fusion	0.8732	0.8350	0.8324
Logistic Regression	0.7686	0.7132	0.6597

The pattern is stable across all three metrics, and the ordering has a mechanistic reading rather than being a numerical accident. Tree-based ensembles lead because the combined representation can be seen as a table with 128 marginal statistical summaries, while axis-aligned splits enable exploiting marginal monotonicity without modelling interaction terms. XGBoost slightly beats the random forest, achieving 0.033 higher macro-F1, thanks to the boosting technique being able to fit additional trees on the class confusion residual, which plays a big role especially for the minority classes, which would be underrepresented by the bagging technique. The MLP falls just short of the two winning models, as expected by the number of samples: for 4,512 training windows and 128 features, a 128-by-64 network has too many parameters relative to the number of observations and causes an overfitting of 0.065 macro-F1. The logistic regression is far behind with 0.6597 macro-F1, although the difference in balanced accuracy is small.

The two-tower fusion architecture is the informative case. Its balanced accuracy of 0.8350 matches the random forest exactly, yet its macro-F1 of 0.8324 is 0.045 below it. Balanced accuracy and macro-F1 differ only in how precision enters, so the pattern indicates that the two-tower model recovers the minority classes about as often as the random forest but pays a precision penalty when it does. Explicit modality separation therefore preserves useful signal and keeps the model lightweight, but it does not by itself beat a well-tuned tree ensemble on this representation and at this sample size. We report that outcome as measured rather than framing the architecture as the best option.

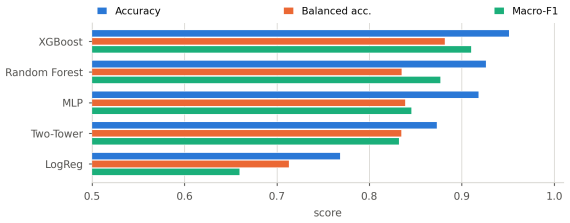


Figure 2: Detector ranking across the three evaluation metrics.

4.2 Ablation Study

A ranking of detectors on the fused input does not by itself show that fusion is responsible for the performance. The fused input could be carried entirely by one modality, in which case the second modality adds cost without adding information. To test this directly, we retrain each tabular detector on three input sets under the identical split, namely EEG only with 96 features, peripheral only with 32 features, and the fused 128 features. Table 4 reports the outcome.

Fusion helps for every detector, and the improvement is statistically significant in all four cases after the exact McNemar test on paired per-window correctness. The result therefore answers the question rather than assuming it. The magnitude of the gain varies in an interpretable way. Logistic regression benefits most at $+0.186$, which is expected because a linear model gains the most from concatenating two views that are each only partially informative, since it cannot construct an interaction between them on its own. The tree ensembles gain less, between $+0.070$ and $+0.093$, because they can already model inter-

Table 4: Modality ablation under the identical hold-out.

Model	Better single	Single	Fused	Δ	McNemar p
XGBoost	Peripheral	0.841	0.910	+0.070	$< 10^{-4}$
Random Forest	EEG	0.784	0.877	+0.093	$< 10^{-4}$
MLP	Peripheral	0.799	0.846	+0.047	0.004
Logistic Regression	Peripheral	0.475	0.661	+0.186	$< 10^{-4}$

actions within whichever single view they receive, so the second view contributes genuinely new information rather than recoverable structure. The MLP gains the least at +0.047 with the weakest evidence, at $p = 0.004$ but the smallest effect, which is consistent with the capacity explanation given above.

A further and less obvious observation is that the better single modality is not the same for every detector. The random forest does better on EEG alone, whereas XGBoost, the MLP, and logistic regression do better on the peripheral channels alone. This is worth stating because it means no single modality is uniformly dominant, and a study that tested fusion against only one of the two single-modality baselines could report a misleading conclusion in either direction. Figure 3 plots the gain as a dumbbell, so that the distance between the two markers can be read directly as the size of the effect that the delta column reports.

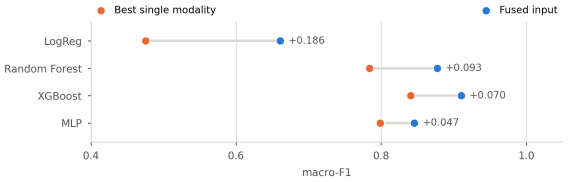


Figure 3: Fusion gain over the best single modality for each detector.

4.3 Class-Level Analysis

We inspect the selected detector at class level to understand which task states are recovered cleanly and where the remaining errors concentrate. Figure 4 shows how the selected XGBoost model distributes its errors on the same stratified 80/20 test split, and Table 5 provides the corresponding class-level precision, recall, and F1-score.

The diagonal of the confusion matrix is dominant, confirming that most task states are recovered reliably, while the remaining off-diagonal mass is concentrated in a small number of confusions among semantically close states. The per-class table explains the aggregate macro-F1 reported above and clarifies the ordering in a way the scalar score cannot.

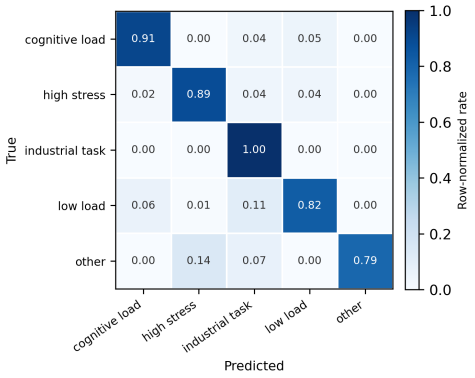


Figure 4: Row-normalized confusion matrix for the selected XGBoost detector.

Table 5: Per-class metrics for the selected XGBoost detector.

Class	Precision	Recall	F1-score
Cognitive load	0.9368	0.9128	0.9247
High stress	0.9091	0.8889	0.8989
Industrial task	0.9635	0.9972	0.9800
Low load	0.9167	0.8250	0.8684
Other	1.0000	0.7857	0.8800

Industrial task is recovered almost perfectly, at 0.9972 recall, which is expected given that it is both the majority class and the most distinctive condition, since the assembly blocks involve sustained physical movement that the peripheral channels capture directly. Low load is the most difficult state, at 0.8250 recall, and the confusion matrix shows where its errors go: 11 percent of low-load windows are predicted as industrial task and 6 percent as cognitive load. Both confusions are physically plausible, as a rest phase immediately following an intensive block exhibits residual autonomic arousal, while 60-second windowing with 50 percent overlap means transitional windows straddle condition boundaries. Conversely, the residual class displays high precision (1.0000) alongside a lower recall of 0.7857: predictions for this class are strictly accurate when made, but roughly one in five true residual windows is missed. This pattern reflects the strong class penalty assigned during training to compensate for its small

sample size. High stress sits between these extremes at 0.8889 recall, confused mainly with cognitive load, which is consistent with shared sympathetic activation across both conditions.

4.4 Real-Time Feasibility

An assistive robot must react within the timescale of the interaction, so a detector that is accurate but slow does not answer the deployment question. Table 6 reports the median per-window inference cost of each detector on CPU, measured over repeated timed batches after warm-up, and Figure 5 shows the same costs on a logarithmic scale so that detectors two orders of magnitude apart stay legible side by side. Logistic regression is the cheapest at about 2.1 microseconds per window, the two-tower model follows at roughly 2.4 microseconds, the multilayer perceptron costs 4.9 microseconds, XGBoost costs about 11 microseconds, and the random forest is the most expensive at approximately 232 microseconds, which is about 111 times the cost of the cheapest detector. The spread is therefore wide enough to matter for a deployment even though every detector is fast in absolute terms, and the cost ordering does not follow the accuracy ordering, because the cheapest model is the weakest on macro-F1 and the most expensive is only the second strongest. A deployment therefore has to trade one against the other rather than take the top of either list.

Table 6: Median per-window inference cost on CPU for each detector.

Detector	Latency (μs)	Throughput (win/s)
Logistic Regression	2.1	479,048
Two-Tower	2.4	412,159
MLP	4.9	203,033
XGBoost	11.4	87,760
Random Forest	232	4,317

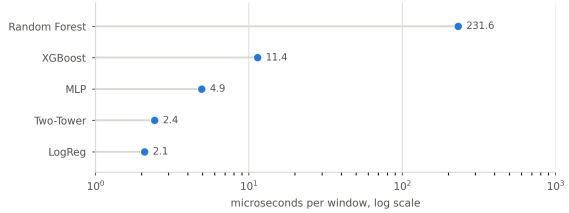


Figure 5: Median inference cost per window for each detector.

5 CONCLUSION

Adaptive HRC relies on continuously recognizing the operator's task and mental state, such as cognitive workload, acute stress, and physical execution, to provide context-aware assistance. However, decoding these states in industrial environments is challenging due to high inter-subject variability, motion artifacts, differing dynamics between central (EEG) and peripheral (EDA, ECG, EMG) sensors, and severe class imbalance.

To address these issues, this paper proposed a compact multimodal pipeline combining 128 fused physiological features with class-aware learning across five task conditions. Results confirm that multimodal fusion significantly enhances task discrimination, with XGBoost delivering the strongest performance under microsecond-level latency.

A modality ablation under the identical protocol shows that the fused input improves macro-F1 over the better single modality for every detector, by between 0.047 and 0.186, and that no single modality is uniformly dominant across detectors. We further stated the scope of the reported numbers candidly: because the hold-out shares operators between its partitions, the figures are a within-subject ceiling, and a subject-disjoint evaluation of the same features gives a substantially lower result.

This work can be extended to real-life deployment and expanding the available task varieties in the dataset. This will help assess how the system behaves in practical settings and whether it remains effective when exposed to a broader range of physiological patterns and collaborative situations.

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